

Musa Jawad

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SUMMARY

Backend/full-stack software engineer with internship experience building APIs, data-driven web features, and production fixes for user-facing platforms. Strong in **JavaScript/TypeScript, Python, SQL, PHP, React, Node.js, Django, REST APIs, testing, and distributed systems fundamentals**.

EDUCATION

University of Calgary - Bachelor of Science in Software Engineering

Calgary, AB | 2025

EXPERIENCE

Fullstack Developer Intern - Self Justice

Dec 2024 - May 2025

- Built and maintained **backend features for a legal information platform used by 1,500+ users**, working in Agile Scrum with JavaScript, REST APIs, and frontend integrations.
- Developed **API endpoints and validation logic** for legal deadline tracking, helping reduce manual tracking work and improve form completion flow.
- Debugged form-submission and data-processing bottlenecks, **reducing processing time by roughly 50%**.

Backend Developer Intern - Inwit

Jun 2024 - Dec 2024

- Developed backend-driven pages and server-side features in **PHP using MVC patterns** for a web platform serving about 5,000 monthly users.
- Optimized **SQL queries and data access patterns**, reducing page load times by about 35%.
- Automated recurring server-side tasks and improved error handling, reducing manual intervention for the team.

Vice President of Finance - Code the Change YYC

Jan 2023 - Sep 2025

- Managed club finances, reimbursements, expense tracking, and documentation for a student tech nonprofit, maintaining **transparent budget records**.
- Led budgeting and resource planning for Canada-wide hackathon events with **300+ participants**, collaborating with technical and operations teams.

PROJECTS

MediScanner | React, Node.js, Firebase, PaddleOCR, Capacitor, Axios

Jan 2025 - Apr 2026

- Built a web and mobile app that extracts **Drug Identification Numbers (DINs)** from medication images using OCR preprocessing and regex-based validation.
- Integrated the **Canadian Drug Product Database and Perplexity AI** for real-time medication lookup and reduced manual lookup errors.
- Tested the OCR pipeline across **500+ image cases** and implemented authentication, storage, and deployment using Firebase.

Distributed Agar.io Clone | Phaser, TypeScript, Django, SQL, WebSockets

Mar 2026 - Jun 2026

- Built a **real-time multiplayer browser game** with WebSocket-based state synchronization for player movement and game updates.
- Designed a layered backend with a **proxy server, Django service, SQL persistence, and replicated data components** for fault-tolerance concepts.
- Used **Lamport timestamp ordering and replica synchronization** to handle distributed event consistency across backend components.

TECHNICAL SKILLS

- Languages:** JavaScript/TypeScript, Python, Java, SQL, PHP, C/C++, HTML/CSS
- Frameworks:** React, Node.js, Express.js, Django, Flask, FastAPI
- Backend & Architecture:** REST APIs, WebSockets, MVC, layered architecture, distributed systems, SQL
- Tools & Platforms:** Git, Docker, Firebase, Google Cloud Platform, VS Code, IntelliJ IDEA, PyCharm
- Testing & Libraries:** JUnit, PHPUnit, Pandas, NumPy, Matplotlib